Infinite Sequences and Series

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Limits of Sequences

A **sequence** is a list of numbers in a definite order:

$$\{a_n\}: \quad a_1,\ a_2,\ a_3,\ \ldots,\ a_n,\ \ldots.$$

In the above sequence, a_n is the *nth term* of the sequence. A sequence $\{a_n\}$ has the **limit** L, and we write

$$\lim_{n\to\infty} a_n = L,$$

if we can make the terms a_n as close to L as we like by taking n sufficiently large. If this limit exists, we say that the sequence **converges**; otherwise, the sequence **diverges**.

Example. Sequences are often defined by giving a formula for the nth term. For instance, consider the sequence $\{a_n\}$ whose nth term is given by

$$a_n = \frac{n}{n+1}: \qquad \frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \frac{4}{5},\ \ldots.$$

We check to see if the sequence is convergent: dividing each term throughout by the highest power, we have

$$\lim_{n\to\infty} \frac{n}{n+1} = \lim_{n\to\infty} \frac{1}{1+1/n} = \frac{1}{1+0} = 1.$$

Thus, the sequence is convergent, and its limit is 1.

Example. Consider the sequence $a_n = (-1)^n$. Observe that writing out the terms in this sequence, we have

$$-1,\ 1,\ -1,\ 1,\ -1,\ \ldots.$$

Thus, a_n does not approach any number, and the limit of the sequence does not exist: that is, $\{a_n\}$ is divergent.

Let us investigate the convergence of $\{a_n\}$, where

$$a_n = \frac{\ln n}{n}.$$

Both the numerator and the denominator approach infinity as $n \to \infty$. Note, however, that we *cannot* directly use l'Hôpital's Rule because it only applies to *functions*, and not sequences. We instead consider the function $f(x) = \ln x/x$: then,

$$\lim_{x\to\infty} \frac{\ln x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = 0.$$

Thus, the sequence converges to 0.

Note that the definition of the limit for sequences is very similar to the one we used for functions. The following result thus becomes apparent.

Theorem 1. *Let $\{a_n\}$ be a sequence, and let $f(x)$ be a function such that $f(n) = a_n$ when n is an integer. If L is the limit of $f(x)$ as $x \to \infty$, then L is the limit of the sequence as well; that is,*

$$\lim_{x\to\infty} f(x) = L \quad \Longrightarrow \quad \lim_{n\to\infty} a_n = L.$$

We now list down some laws for limits of sequences.

Proposition 2. *Let $\{a_n\}$ and $\{b_n\}$ be convergent sequences, and let k be some constant. Then,*

- $\lim\limits_{n\to\infty} (a_n + b_n) = \lim\limits_{n\to\infty} a_n + \lim\limits_{n\to\infty} b_n$
- $\lim\limits_{n\to\infty} ka_n = k \lim\limits_{n\to\infty} a_n$
- $\lim\limits_{n\to\infty} (a_n b_n) = \lim\limits_{n\to\infty} a_n \cdot \lim\limits_{n\to\infty} b_n$
- $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = \dfrac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$ *if* $\lim\limits_{n\to\infty} b_n \neq 0$

We also have the following helpful results:

- *If* $\lim\limits_{n\to\infty} |a_n| = 0$, *then* $\lim\limits_{n\to\infty} a_n = 0$.
- *If* $\lim\limits_{n\to\infty} a_n = L$, *and $f(x)$ is a function (continuous at L), then*

$$\lim_{n\to\infty} f(a_n) = f(L).$$

Example. Let us investigate the convergence of the sequence

$$a_n = \frac{n!}{n^n}, \text{ where } n! = 1 \cdot 2 \cdot 3 \cdot \cdots \cdot n.$$

We first write out the sequence:

$$a_1 = \frac{1}{1}, \quad a_2 = \frac{1 \cdot 2}{2 \cdot 2}, \quad a_3 = \frac{1 \cdot 2 \cdot 3}{3 \cdot 3 \cdot 3}, \quad \ldots.$$

More generally, we have

$$a_n = \frac{1 \cdot 2 \cdot 3 \cdot \cdots \cdot n}{n \cdot n \cdot n \cdot \cdots \cdot n} = \frac{1}{n}\left(\frac{2 \cdot 3 \cdot \cdots \cdot n}{n \cdot n \cdot \cdots \cdot n}\right).$$

The key observation to make here is that

$$\frac{2 \cdot 3 \cdot \cdots \cdot n}{n \cdot n \cdot \cdots \cdot n} \leq 1,$$

since the denominator must be greater than or equal to the numerator. Hence,

$$0 \leq a_n \leq \frac{1}{n} \cdot 1.$$

We know that $1/n$ approaches 0 as n goes to infinity: so, taking limits, we find that

$$0 \leq \lim_{n\to\infty} a_n \leq 0.$$

By the Squeeze Theorem, we conclude that

$$\lim_{n\to\infty} a_n = 0.$$

Series

If we add the terms of an infinite sequence $\{a_n\}$, we obtain a **series**:

$$\sum_{n=1}^{\infty} = a_1 + a_2 + \cdots + a_n + \cdots .$$

Given this series, we can consider its **partial sum**

$$s_n = \sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n.$$

We can then consider the sequence of partial sums: if this sequence is convergent with

$$\lim_{n\to\infty} (s_n) = s,$$

then the infinite series is **convergent**, and we write

$$\sum_{n=1}^{\infty} a_n = s,$$

where s is the **sum** of the series. Otherwise, the series called **divergent**.

Example. The *geometric series* is given by

$$a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1},$$

where $a \neq 0$. We obtain each term from the preceding one by multiplying it by the *common ratio* r. If $r = 1$, the partial sum s_n is given by

$$s_n = a + a + a + \cdots + a = na.$$

If $r \neq 1$, the partial sum s_n is given by

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a\,(1 - r^n)}{1 - r}.$$

This allows us to determine when the series converges: if $-1 < r < 1$, we know that $r^n \to 0$ as $n \to \infty$, and so

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} \frac{a\,(1 - r^n)}{1 - r} = \frac{a}{1 - r}.$$

We can thus conclude that the gometric series converges when $|r| < 1$, and diverges when $|r| > 1$.

Exercise. Evaluate the series $\displaystyle\sum_{n=1}^{\infty} \frac{2^{2n-3}}{6^n}$.

Solution. We simplify the series

$$\sum_{n=1}^{\infty} \frac{2^{2n-3}}{6^n} = \sum_{n=1}^{\infty} \frac{2^{2n-3}}{2^n 3^n} = \sum_{n=1}^{\infty} \frac{2^{n-3}}{3^n} = \sum_{n=1}^{\infty} \frac{1}{8} \cdot \left(\frac{2}{3}\right)^n = \frac{1}{12} + \frac{1}{12} \cdot \frac{2}{3} + \cdots$$

This is simply a geometric series with first term $a = 1/12$ and common ratio $r = 2/3$. This converges to

$$\frac{a}{1 - r} = \frac{\frac{1}{12}}{1 - \frac{2}{3}} = \frac{1}{12} \cdot 3 = \frac{1}{4}.$$

Exercise. Does the series $\displaystyle\sum_{k=1}^{\infty} \frac{1}{\sqrt{k+3}+\sqrt{k+4}}$ converge or diverge?

Solution. We begin by rationalizing the summand

$$\begin{aligned}\frac{1}{\sqrt{k+3}+\sqrt{k+4}} &= \frac{1}{\sqrt{k+3}+\sqrt{k+4}} \cdot \frac{\sqrt{k+3}-\sqrt{k+4}}{\sqrt{k+3}-\sqrt{k+4}} \\ &= \frac{\sqrt{k+3}-\sqrt{k+4}}{k+3-k-4} = \sqrt{k+4}-\sqrt{k+3}.\end{aligned}$$

Observe that this is *not* a geometric series, so we consider the limit of the partial sums. Expanding this expression, we find that it is a *telescoping sum*:

$$\begin{aligned}s_n &= \sum_{k=1}^{n} \frac{1}{\sqrt{k+3}+\sqrt{k+4}} = \sum_{k=1}^{n} \sqrt{k+4}-\sqrt{k+3} \\ &= \left(\sqrt{5}-\sqrt{4}\right)+\left(\sqrt{6}-\sqrt{5}\right)+\cdots+\left(\sqrt{n+4}-\sqrt{n+3}\right) \\ &= \sqrt{n+4}-\sqrt{4}.\end{aligned}$$

The limit of the partial sums is

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty}\left(\sqrt{n+4}-\sqrt{4}\right) = \left(\lim_{n\to\infty}\sqrt{n+4}\right)-2,$$

which approaches infinity; hence, the series diverges.

Power Series

We now explore functions given by series. A **power series about $x = a$** is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n + \cdots$$

in which the **center** a and the **coefficients** $c_0, c_1, c_2, \ldots, c_n, \ldots$ are constants. For each fixed x, the power series is a series of constants that we can test for convergence or divergence—a power series may converge for some values of x and diverge for other values of x. The sum of the series is a function

$$f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n + \cdots$$

whose domain is the set of all x for which the series converges. For the purposes of our discussion, let us assume that our power series is always centered at $a = 0$, such that our series has the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots$$

Example. Consider the power series with coefficients $c_n = 1$ for all n. We obtain the geometric series

$$\sum_{n=0}^{\infty} 1 \cdot x^n = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots,$$

which converges when $|x| < 1$. In particular, we have a power series representation of a function:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1.$$

We can do *calculus* on power series: we simply differentiate or integrate term by term, and the radius of convergence of the power series remains the same.

Example. Let us try to determine a power series representation for $\ln(1+x)$. We recall that

$$\frac{d}{dx}\ln(1+x) = \frac{1}{1+x} = \frac{1}{1-(-x)}.$$

We can thus use the geometric series:

$$\frac{1}{1-(-x)} = 1 - x + x^2 - x^3 + \cdots, \quad |x| < 1.$$

Integrating both sides of this equation yields the desired power series:

$$\ln(1+x) = \int 1 - x + x^2 - x^3 + \cdots \, dx = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + c.$$

To determine the value of c, observe that plugging in $x = 0$ yields

$$\ln(1+0) = c \quad \Longrightarrow \quad c = 0.$$

Hence, we conclude that

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \quad |x| < 1.$$

Question. *Which functions have representations as power series? How can we find such representations?*

Suppose that a function $f(x)$ can be represented by a power series

$$f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \cdots.$$

Let us try to determine what the coefficients c_n must be.

- Observe that, plugging in $x = 0$, we have $f(0) = c_0$.
- If we differentiate the series, we have

  $$f'(x) = c_1 + 2 \cdot c_2 x + 3 \cdot c_3 x^2 + 4 \cdot c_4 x^3 + \cdots.$$

  Plugging in $x = 0$, we have $f'(0) = c_1$.
- If we differentiate again, we have

  $$f''(x) = 2 \cdot c_2 + 3 \cdot 2 c_3 x + 4 \cdot 3 c_4 x^2 + \cdots.$$

  Plugging in $x = 0$, we have $f''(0) = 2 \cdot c_2$, or $c_2 = f''(0)/2$.
- Differentiating yet again, we have

  $$f'''(x) = 3 \cdot 2 \cdot c_3 + 4 \cdot 3 \cdot 2 \cdot c_4 x + \cdots$$

  Plugging in $x = 0$, we have $f'''(0) = 3 \cdot 2 \cdot c_3$, or $c_3 = f'''(0)/3!$.

We can perhaps spot the pattern: if we repeatedly differentiate and substitute $x = 0$, we find that

$$f^{(n)}(0) = n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot 3 \cdot 2 \cdot c_n \quad \Longrightarrow \quad c_n = \frac{f^{(n)}(0)}{n!}.$$

We can thus conclude: if f has a power series expansion (centered at 0), it must be of the form

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \frac{f'''(0)}{3!} x^3 + \cdots.$$

This infinite series is called the **Maclaurin series** of $f(x)$, named after the 18th Century Scottish mathematician Colin Maclaurin, who popularized the idea in his calculus book *Treatise of Fluxions* (1742).

Example. Let us find the Maclaurin series of the function $f(x) = e^x$. Its derivatives are fairly predictable: for all n,

$$f^{(n)}(x) = e^x \quad \Longrightarrow \quad f^{(n)}(0) = 1.$$

Hence,

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n!} + \cdots.$$

A Summary of Important Maclaurin Series

The following list contains some important Maclaurin series and the values of x over which they converge.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots \quad \text{for } |x| < 1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad \text{for } |x| < 1$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad \text{for all } x \in \mathbb{R}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = \frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \quad \text{for all } x \in \mathbb{R}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \quad \text{for all } x \in \mathbb{R}$$

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!} x^2 + \cdots \quad \text{for } |x| < 1$$

Extra Notes on Infinite Sequences and Series

Power Series for Sine and Cosine. We determine the Maclaurin series for some trigonometric functions.

Let $f(x) = \sin x$, and so $f(0) = 1$. Observe that

$$\begin{aligned} f'(x) &= \cos x & \Longrightarrow \quad f'(0) &= 1 \\ f''(x) &= -\sin x & \Longrightarrow \quad f''(0) &= 0 \\ f'''(x) &= -\cos x & \Longrightarrow \quad f'''(0) &= -1 \\ f^{(4)}(x) &= \sin x & \Longrightarrow \quad f^{(4)}(0) &= 0 \\ &\vdots & & \vdots \end{aligned}$$

We find that the derivatives repeat in cycles; moreover, only the odd derivatives are non-zero. Hence,

$$\begin{aligned} f(x) &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots \\ &= \frac{1}{1!}x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}. \end{aligned}$$

- Let $f(x) = \cos x$. We may compute this directly, or we can simply differentiate the Maclaurin series for $\sin x$:

$$\begin{aligned} \cos x &= \frac{d}{dx}\sin x = \frac{d}{dx}\left(\frac{1}{1!}x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \cdots\right) \\ &= 1 - \frac{3}{3!}x^2 + \frac{5}{5!}x^4 - \frac{7}{7!}x^6 + \cdots \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}. \end{aligned}$$

Binomial Series. Let us determine the Maclaurin series for

$$f(x) = (1+x)^k, \text{ where } k \in \mathbb{R}.$$

First, observe that

$$\begin{aligned} f'(x) &= k(1+x)^{k-1} \\ f''(x) &= k(k-1)(1+x)^{k-2} \\ &\vdots \\ f^{(n)}(x) &= k(k-1)\cdots(k-n+1)(1+x)^{k-n}. \end{aligned}$$

Thus, with $f(0) = 1$, we have

$$\begin{aligned} f'(0) &= k \\ f''(0) &= k(k-1) \\ &\vdots \\ f^{(n)}(0) &= k(k-1)\cdots(k-n+1), \end{aligned}$$

and we have the Maclaurin series

$$(1+x)^k = \sum_{n=0}^{\infty} \frac{k(k-1)\cdots(k-n+1)}{n!}x^n.$$

This series is called the **binomial series**. The traditional notation for the coefficients in the binomial series, called *binomial coefficients*, is

$$\binom{k}{n} = \frac{k\,(k-1)\,(k-2)\cdots(k-n+1)}{n!}.$$

Observe that if k is a nonnegative integer, then the terms are eventually 0 and the series is finite.